MutantXL closure at scale: degree 4 holds to N = 42 at m = 4, and rises past 4 at m = 3, ℓ = 7

These are the results of `research/dreg_closure_scale_20261016/`. The full text is `RESULTS.md`, and the run report is `reports/run-20261016T1705Z/`. **Class:** stage diagnostic. The `n = 131` rows are **derived extrapolations**.

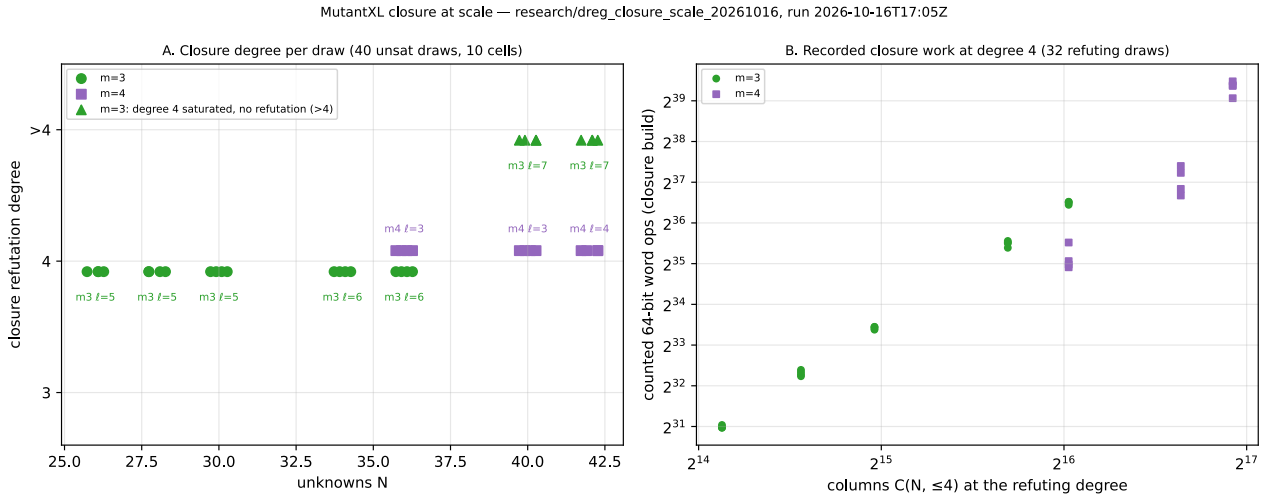


Figure 1: 40 unsatisfiable draws, four per cell, run 2026-10-16T17:05Z.

Prediction	outcome	confidence	evidence
Q4: the closure refutes at ≤ 4 in every draw	**refuted**: 0 of 8 at m = 3, ℓ = 7; 32 of 32 elsewhere	about 55%	`score-output.txt`
Q5: XL never refutes at 4	**holds**: 0 of 40	high	same
Q6: $\log_2$ ops ÷ $\log_2$ cols in 2.0–2.4	**holds**: 2.18–2.33	—	same

- **Registered verdict:** "rises at N = 40".
- **Descriptive reading (not registered):**
  - the rise tracks ℓ rather than N, because m = 4 at N = 40 and 42 stays at 4;
  - the refuting cells have m·ℓ ≤ 18 and the rising cells have 21.
- **Soundness control:** the one registered satisfiable draw was not refuted. That is thin, and the report says so.

ECC2K-130 (derived)

m	degree	w	ℓ	$\log_2$ total	vs rho
3	4, constant — refuted at ℓ = 7	1	40	82.27	+21.46
3	5, constant	1	41	85.18	+24.37
4	4, constant — holds to ℓ = 4	1	33	68.43	+7.62
4	5, constant	1	34	71.13	+10.32

- No row beats rho, and the units are unconverted.
- **Correction 2026-10-30.** The m = 4 degree-5 row (2^71.13, +10.32) is priced at w = 1, a floor (lower bound), not an optimistic estimate. At the fitted w (degree 5: 2.83, 95% CI 2.59 to 2.92) it is **+61.45 bits over rho in rho-step units** (bootstrap 55.8 to 64.0; degree-4 proxy +52.21), and a structured / block-Wiedemann elimination prices it at about +43. The interval excludes the extrapolation from about 2^16 to 2^36 columns. Source: `research/m4_matrix_profile_20261030/README.md`; `RESEARCH_ECC2K130_DESCENT_DEGREE.md` §7 correction.
- **The next cell** is m = 4, ℓ = 5 at degree 4 (about 179k columns). It is at this container's memory limit.
- **Degree 5 at N ≥ 40** needs a sparse closure.